

BIOT 753 & BIOT 853

CELL CULTURE

Instructor

Dr. Kristen Johnson

Office Hours: Mondays, 1-2pm

WHERE/WHEN

- **Lecture (RM 345): TU/TR, 10:30-12pm**
- **Lab (650EC): 1-3:30pm or 6:30-9pm, TU/TR**

COURSE DESCRIPTION

Principles and technical skills fundamental to the culture of animal and plant cells, tissues, and organs. Introduction to the techniques of sub-culturing, establishing primary cultures, karyotyping, serum testing, cloning, growth curves, cryopreservation, hybridoma formation and monoclonal antibody production, and organ cultures. Application of cell culture to contemporary research in the biological sciences.

Course Objectives. We aim to teach:

1. The history, theory, and biology of mammalian cell culture.
2. Biotechnology and basic research applications of eukaryotic (and prokaryotic) cell culture.
3. Aseptic technique and cell culture assays important to basic research and biotechnology.
4. Science communication through oral presentations, grant/project proposal writing, and lab reports.
5. Critical thinking and problem solving through evaluation of real data sets incorporating cell culture techniques.

Student Learning Outcomes. After completing the cell culture course, students should be able to:

1. Understand the history, theory and biology behind the cell culture techniques and reagents utilized in the laboratory.
2. Aseptically culture mammalian cells.
3. Learn to use equipment properly to carry out a variety of cell culture techniques, including cell subculturing, viability, cytotoxicity, cryopreservation, transfection, immunofluorescence, flow cytometry, etc.
4. Critically evaluate data from primary research articles to understand the application of classroom learning to current basic research fields.
5. Act like researchers by developing a project plan, writing a grant proposal, carrying out experiments, statistically analyzing data, using graphing software to display data appropriately, and present findings to an audience.

REQUIRED MATERIALS

COURSE WEBPAGE

- *Online Textbook*
- MyCourses modules will contain: Power Point presentations, all readings, useful websites, and online assignments.

Instructor

Contact:

kristen.johnson@unh.edu

Office: *Room 630*

LECTURE COURSE SCHEDULE (JOHNSON)

Note: This course schedule is subject to change! All changes will be communicated to students via email/MyCourses.

Week	Date	Lecture
1	Jan 24 Jan 26	Introduction to Course; Terminology, Basic cell culture
2	Jan 30 Feb 2	Biology of Cultured Cells & Cell Characterization
3	Feb 7 Feb 8	Environmental Factors/Nutrition in vitro
4	Feb 14 Feb 16	Quantifying Cell Growth and Other Procedures Quiz #1
5	Feb 21 Feb 23	Studying Cancer Cells: Identifying cell growth and attachment problems, etc.
6	Feb 28 Mar 2	
7	Mar 7 Mar 9	MIDTERM EXAM
8	Mar 21 Mar 23	Primary Culture
9	Mar 28 Mar 30	Stem Cell Culture & Differentiation
10	Apr 4 Apr 6	Gene Delivery & Cloning Quiz #2
11	Apr 11 Apr 13	Mammalian Cell Culture Applications <i>Site visit to Pfizer/Lonza (TBD)</i>
12	Apr 18 Apr 20	Cutting Edge approaches in Cell Culture
13	Apr 25 Apr 27	<i>Flex days (Snow Day accommodation)</i>
14	May 1 May 3	Grad Student Independent Project Presentations
15	May 9	FINAL EXAM

Methods of Testing/Evaluation: The lecture course grade will be based on following:

Method of Evaluation	Total Point Value
2 Exams* – 150 pts ea	300
Weekly in-class or homework assessments – 15 pts ea	150
2 Quizzes – 75 pts ea	150
TOTAL POINTS AVAILABLE	600

**Note: Additional questions will be added to each exam for 800/graduate-level students.*

For graduate students, your overall BIOT 853 grade will be determined by averaging the lecture and lab grades (50%/50%).

Course Assessments

- **Exams:** A midterm exam (March 7th) and a final exam (May 9th) will be given for this course. Both exams will cover material from lecture and the background and protocols from experiments carried out in lab. The format of these exams will largely be short answer/essay, and questions will be like those in the weekly homework assignments.
- **Quizzes:** Two 45-minute quizzes will be given during the semester - during Thursday lecture periods. Both quizzes will cover material from lecture **and** the experiments carried out in lab. These quizzes will contain a variety of formats: fill-in-the-blank, matching, true-false, multiple choice, and short answer.
- **Weekly Homework Questions / In-class Assignments:** To support the application of the content learned through lectures, you will be assigned either short answer questions online (posted Tuesday pm or Wednesday am) pertaining to the week's lecture and reading (and sometimes lab material) or work on individual or collaborative assignments during Thursday lecture classes. For online assignments, questions are due each week by **Sunday, 11:59 pm. Late assignments will not be accepted.**

THE DO's and DON'Ts...The Fine Print

Study and time spent outside of class:

While the actual study time needed to succeed will vary among individual students, on average students should expect to put in 3 hours of study per every hour spent in class or laboratory. Work outside of class includes but is not limited to taking notes on readings, reviewing class notes/Power Points, reviewing online video content, reviewing, and preparing lab materials and assignments, and preparing for and completing online homework assignments/posting on discussion boards. Be sure to review the material/ your notes daily rather than cramming it into one time. **Be proactive rather than reactive!**

GENERAL COURSE POLICIES

Attendance

- Please arrive on time. Do not leave early.
- Regular attendance, particularly in lab, is vital to your success in this course! While occasional circumstances, over which you have no control, may necessitate absence, the content presented in the activities missed by you is a segment of a broad base of information being taught. For this reason, your attendance is expected.
- Attendance will be noted during every lab and lecture sections. If you miss a lecture class, please contact me (by email in advance) to let me know. **Because this course is so lab intensive, two or more lab absences will result in a 5% reduction in your grade.**

- It is your responsibility to get all lecture and lab material from a classmate to make up what you have missed during any absence. Also, be sure to check MyCourses/Canvas for relevant Power Points, handouts, assignments, etc.

Lab/Classroom Conduct

- *Laboratory Conduct:* You must wear long pants or a long skirt, wear close-toed shoes, and pull any hair back in the cell culture lab. You may not chew gum, bring your cell phone, or have any electronic devices in the lab. You will be given a dedicated lab coat and goggles/glasses that will be kept in the cell culture room, as well as a dedicated lab notebook that will be provided for you. *You will be working in an active research lab, and all sterile procedures as well as BSL-2 protocols will be strictly enforced!*
- *Cell phones:* Please **turn off/silence** and put away your cell phone before you enter the classroom. Cell phones are NOT permitted in the cell culture room.
- *Computer Use:* You may use a laptop *during lecture* for taking notes ONLY. You should NOT be using your computer for social networking, emailing, playing games, etc. Such behavior distracts you and your classmates from the lecture and is considered disrespectful to the professor and your classmates. If you are judged to be misusing your computer in any way, you will be asked to put it away for the remainder of the lecture.
- *Food:* There is a strict policy of NO food or drinks in the laboratory. However, you may choose to have drinks in the classroom.

Make-up Quizzes/Exams

- **Exams and quizzes are given only on the announced date.** Make-up exams and quizzes will be given ONLY if it is missed for a valid (documented) reason and discussed ahead of time.

Academic Honesty

The Biology/Biotechnology Programs at UNH Manchester will strictly adhere to the [University policy on academic honesty](#), as published in the UNH Student Rights, Rules, and Responsibilities Handbook. It is imperative that you review this policy in order to understand the rules and responsibilities with regard to honest, academic integrity, plagiarism and cheating. By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual homework assignments or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report. *Your assignments submitted through Canvas will be routinely screened for plagiarism using sophisticated software.* It is strongly recommended that you complete the [Tutorial on Plagiarism in order to fully understand the definition\(s\) and all forms of it.](#)

Cheating is mainly concerned with copying on quizzes, exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and placed upside down on the table in front of you for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

UNH Manchester Campus Support

Library: The UNH Manchester librarians are available to assist you with your research. You can contact a librarian by calling 603-641-4173 or by emailing unhm.library@unh.edu.

The following online resources provide information about library resources and services:

- UNH Manchester Library webpage: <https://manchester.unh.edu/library>
- Online Research Guides: <https://libraryguides.unh.edu/index.php?b=s>
- Access Library Resources Remotely: <https://libraryguides.unh.edu/remotearchive>
- Reserve a study room for Zoom classes: <https://libraryguides.unh.edu/remotearchive/studyrooms>

Center for Academic Enrichment: Provides academic support services, including individual tutoring, for all students. Make an appointment through the CAE My Courses site on your Canvas dashboard.

Phone: (603) 641-4113. Email: unhm.cae@unh.edu

Accessibility Services

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Jenessa Zurek. Please reach out to the SAS office via email at jenessa.zurek@unh.edu for registration information and disability related questions. Jenessa Zurek is available through phone and email Mondays and Wednesdays from 9am-2pm.

Mental Health and Wellness

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers free mental health sessions for students. Students can schedule virtual counseling sessions by emailing unhm.advising@unh.edu. Counselors will be available virtually on Monday, Tuesday, and Thursday from 9am-5pm.

The National Suicide Prevention Lifeline provides 24/7, free and confidential support via phone or chat for people in distress, resources for you or your loved ones, and best practices for professionals. Call (800) 273-TALK (8255).

Sexual Harassment and Rape Prevention Program (SHARPP): Provides free and confidential advocacy and direct services to survivors. (<https://www.unh.edu/sharpp>.)

Food Pantry: The campus food pantry has pre-packed bags of non-perishable food items for students in room 437.

Pandemic Related Policies

Mask wearing in class and campus buildings is currently optional. However, please be advised that if this policy changes during the semester, you will be required to adhere to the most current policy.

1) Each classroom entrance is equipped with hand sanitizer and surface wipes.

- **Use hand sanitizer, as you deem appropriate.**
- **Wipe down your personal space/desk prior to class and throw the used wipes away on the way out of class or take them with you.**

2) UNH has developed "Wildcat Pass," a web and mobile app to help each of us keep track of the requirements for following necessary testing, isolation and quarantine rules that will help to keep our community healthy.

- Log into your Wildcat Pass each day.
- Be prepared to show your mobile device or a computer printout of your daily Wildcat Pass upon entry to the building and in class if requested.